AI DRIVEN SELF HELP GROUP





SRI SAIRAM ENGINEERING COLLEGE , TAMBARAM

PROBLEM STATEMENT

PROBLEM



Women SHGs don’t have easy digital tools to manage their money, loans, and savings, and they also find it difficult to sell their products beyond their local area.

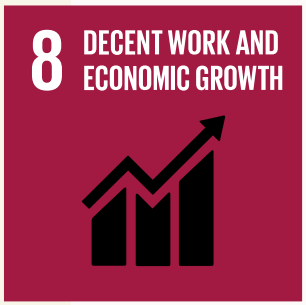
IMPACT



Because of this, women face financial losses, have less income opportunities, and there is lack of transparency, which slows their growth and empowerment

SDGs and TARGETs







SDG 8 – Decent Work and Economic Growth

- Target 8.3: Promote small business growth and entrepreneurship
- Target 8.5: Achieve productive employment and decent work for all

SDG 5 – Gender Equality

- Target 5.1: End all forms of discrimination against all women and girls everywhere
- Target 5.a: Give women equal rights to economic resources, financial services, and ownership

SDG 10 – Reduced Inequalities

- Target 10.2: Promote social and economic inclusion

RESEARCH AND REAL - WORLD DATA

WADHWANI FOUNDATION

First wave or second, women continue to be at a disadvantage

Taking Stock

Priyanka Chaturvedi

The second wave of the pandemic has been creating havoc across the country and as the nation prepares to come to terms with tackling shortages, we also have to be mindful of the third wave. However, one thing that has remained constant is that women have been at a far greater disadvantage and have almost been trivialised to the scheme of things and in terms of planning. The pandemic has impacted women - physically, financially, mentally and has had an impact on their education. It has deepened pre-existing inequalities, be it the they are treated in society, their participation in policy and politics or their pay disparity.

In the first lockdown, the Central government had supported families by providing Rs 500 a month for three months, to nearly 300 million women directly via their Jan Dhan accounts. However, in the second wave, since no nationwide lockdown was announced, that safety net has been taken away and there has been no thought on that front by the Centre either. While women and child empowerment is a state subject, states are already grappling with Covid-related financial impact and GST refund delays.

India's Constitution gives equal rights to women in society as men; however, long-standing patriarchal norms have continued to deny women equal access in the society. Now, for some hard facts on the prevalence of unjust differences between the genders- India ranks 131 out of 156 nations on economic participation and opportunity for women. The female workforce participation in India in 2019 stood at merely 20.22 per cent, and is in danger of falling further, due to the crisis. Women earn 35 per cent lesser than men while the global average is 16 per cent. Women, despite being half the nation's population, contribute only 17 per cent to its economic output. A government that promotes Digital India has again forgotten to factor in its women and the digital divide is only getting more defined in this crisis. Males account for 67 per cent of India's online population, this gap is not just leading to girls falling off the education grid but could also be a factor for women getting vaccinated, since appointments are based on access to a portal.

Besides the above challenges, there has also been a rise in child marriages and malnutrition due to the pandemic and school closures. The UN has been more than a two-fold rise in complaints of domestic violence since the lockdown, but the WCD Minister has claimed it as incorrect, without being sensitive enough to the plight of women who may have feared complaining about it since they were locked down in the same house and victim as the accused. Around 10 million of our girls are increasingly at risk of being forced out of school and into child marriage, child labour, sexual exploitation, teenage pregnancy, poverty, trafficking and violence. The circle of violence for such girls is never ending.

Data suggests that more men as compared to women lost their lives due to Covid-19, but there is little discussion or policy in place to support widows and families who were dependent on the men for economic sustenance. We still have not gauged data to understand whether the low or number of women in the Covid data is because of immunity or because of inability to get access to testing and vaccination. Till date, pregnant women have been kept out of the vaccination ambit in India citing lack of data, despite a strong case being made for it in India and globally. It was only recently that lactating and breastfeeding women were allowed access to vaccines. Denying women in this category the right to decide whether they want to be vaccinated or not is yet another exclusionary tactic, as well as robbing them of their agency.

While it sounds good to make speeches from the Red Fort about women and their participation in nation-building, it is totally another matter when it comes to implementation. Until women become a collective force to safeguard their politics and privilege, we will continue to see the exploitation of their sentiments for vote-bank politics of those working power.



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The author is a Rajya Sabha MP & deputy



54 lakh girls given scholarships: WDC

TUESDAY, MARCH 9, 2021

Times News Network

Patna: Women Development Corporation (WDC) managing director Harjot Kaur Bamhrah on Monday said 54 lakh girls studying in state government-run schools were given scholarships amounting to Rs six crore in 2020-21.

Bamhrah was addressing a conference organised by International Growth Centre (IGC) ADRI CONF

and Asian Development Research Institute (ADRI) in collaboration with the WDC to celebrate International Women's Day on Monday. The theme of the conference was 'Women empowerment: Issues, evidences and policy response'. It was jointly inaugurated by Bamhrah and state development



Delegates at a conference organised by ADRI in association with WDC

Sobhani said fertility rate in Bihar has come down from 3.4 to 2. "However, there was a decline in the sex ratio at birth due to the evil-rooted hegol dominance. The government is working towards raising awareness among people to end the menace of female foeticide," he said.

Ajay Ramachandran, professor of economics at University of Western Australia, said female labour force ratio in India is declining. "It is now below 20% for women as compared to 80% for men. Covid-19 has further aggravated the situation. Women lost their jobs in the Covid era as compared to 60% of men due to higher costs for companies in keeping women in their jobs."

Devesh Sekra, secretary (SC-ST welfare department), said only 36% of the women in India could retain their jobs in the Covid era as compared to 60% of men due to higher costs for companies in keeping women in their jobs.

Dr Ashwini Gupta, assistant professor, ADRI, said there is a need to re-examine the kind of growth people want in the country. "Earlier there was jobless growth in India where a lot of women lost out in getting remunerative jobs," she said.

Jyotsana Jha, director, Centre for Budget and Policy Studies, said there was a deficit in consistent towards greater gender equality. She pointed out that there is less preference for boys in areas where the female labour force participation

'Learn and Earn' is the new mantra

Delhi-based NGO Helping Brainz is organising a soap-making workshop for flood victims on Saturday

Meera Suresh



A LAUDABLE MODEL

SELF-HELP GROUP SAVINGS AS OF MARCH 31

- The SHGs of Andhra Pradesh have an average ₹1.7 lakh savings
- This is the highest among all the states and Union Territories
- This is followed by ₹85,000 per SHG in Telangana



- The savings of SHGs of two Telugu states is way higher than the national average of ₹43,940
- Andhra Pradesh has about 10.98 lakh SHGs
- Telangana has about 6.1 lakh SHGs
- SHGs form nearly 60% of the women electorate in AP
- Each group usually consists of 10 women
- The emphasis is not only on economic activities, but also on social issues such as family welfare, childcare, and literacy



Disadvantaged women step out of shadows to take charge of lives

▶From P 1

For years now, Nurude has been trying to avail of a government scheme which gives subsidies to widows to build homes but has not managed to get her application cleared yet.

"There are many women like me who don't want to be a burden on anyone and would like to earn and live alone with their children. These schemes are meant for us, but very few manage to get their benefits," said Nurude.

Several single women who are either widowed, have remained single due to caste or health issues, or were "left" by their husbands as they could not conceive or had given birth to girl children have contested gram panchayat elections from the drought-prone districts of Marathwada where the caste system is still robust and very few females manage to step outside the shadows of their husband, father or brothers.

"A total of 194 women contested in our organization contested the elections and 60 won. This is such a proud moment for us. Most of these women wouldn't even attend gram sabha meetings initially. We encouraged them to attend the meetings and raise their complaints. When these women would start asking questions, the men would ignore them. Taking from these experiences we decided last year to train these women to contest the gram panchayat elections. We even did a mock election at our organisation last year to boost their confidence," said Rani Shelke, Marathwada representative of Coro India, which works as a facilitator with Ekal Mahila Sangathana. The sanghatana, which was started in 2014, now works with more than 10,000 women. Elections in nearly 15,000 gram panchayats were held earlier this month in the state.





Some of the candidates who won gram panchayat elections (right) Sumantbai Nurude from Bhaswadi, Nanded

For Kaushalya Kalnde (32) from Beed district, a widow with two children, it was not an easy ride though. "First, my mother was opposed to my decision to contest. I told her very clearly that I will go ahead. Then, after she agreed, I got threats from men in my own village who asked me to take back my candidature. While other candidates were hurling vowers with non-vegetarian meals, I couldn't even offer women tea from my house as we had only limited sugar," said Kalnde who lost the elections by just three votes. "But I am not dejected. I will continue to attend meetings and ask questions and will contest in the next elections," she said.

Shelke said they with representatives of the panchayat as samitis will ensure details of all single women from these villages are registered. At the moment there is no record of how many single women live in the villages. "Unless a record is created, how will you ensure that the policies are reaching them?" asked Shelke.

ABSTRACT

The AI-Driven Self-Help Group (SHG) Support Platform aims to empower women in rural and semi-urban areas by addressing challenges in financial management, market access, and digital literacy. Traditional manual record-keeping leads to errors, lack of transparency, and inefficient tracking of savings and loans.

This project proposes a smart digital solution that integrates AI with mobile and web applications to automate bookkeeping, predict loan risks, and provide financial insights. It also includes a marketplace to connect SHG products with buyers and a multilingual chatbot for user support.

By improving financial transparency, enabling better decision-making, and increasing income opportunities, the platform supports women's empowerment .

OUR SOLUTION

We propose an AI-powered mobile application designed to support Women Self-Help Groups (SHGs) by simplifying financial management and improving market access.

KEY FEATURES

Smart Bookkeeping

Record savings, loans, and expenses
Auto-generate financial summaries
Simple interface for easy use

SHG Group Management

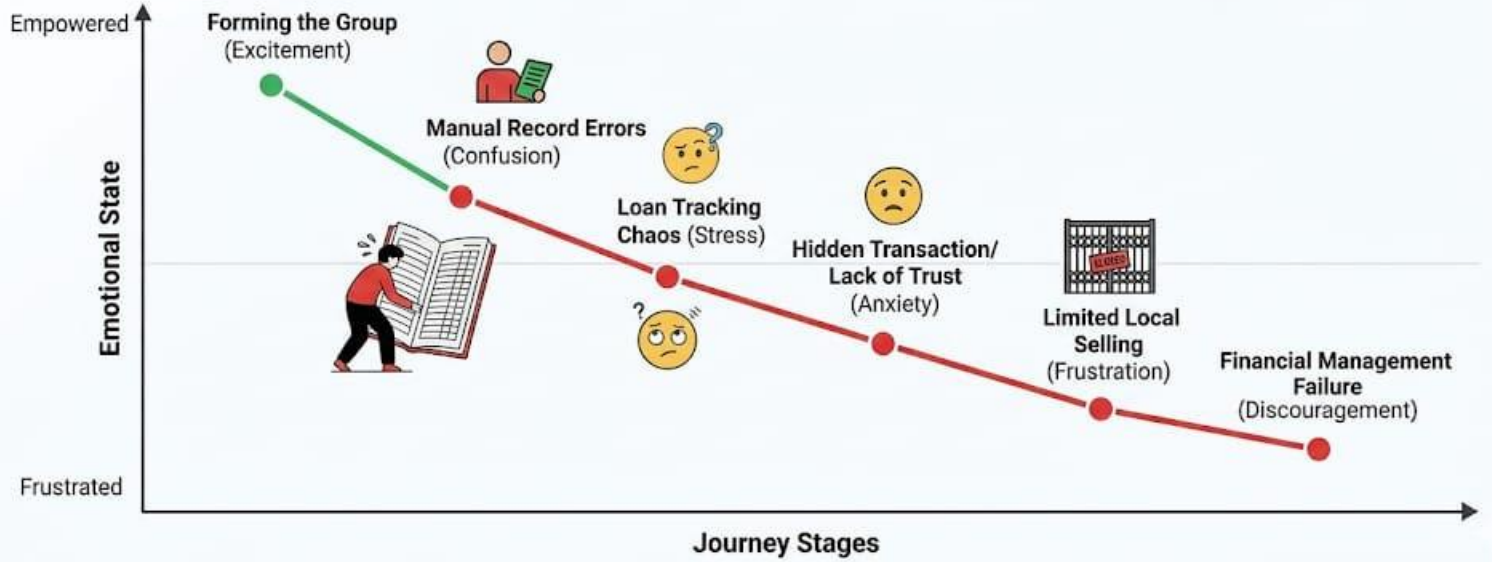
Manage group members
Track individual contributions
Improve transparency

Voice-Based Input

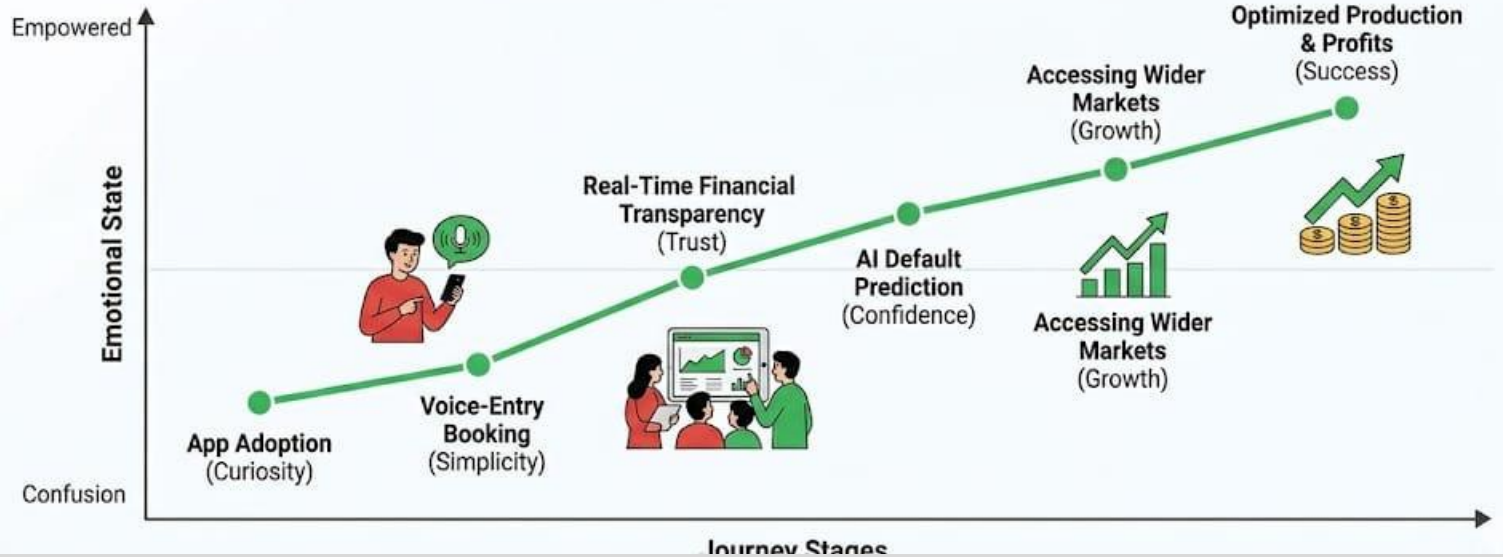
Easy data entry using voice
Useful for low literacy users

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SHG Member's Perspective (The Problem: Manual Struggle) — Before

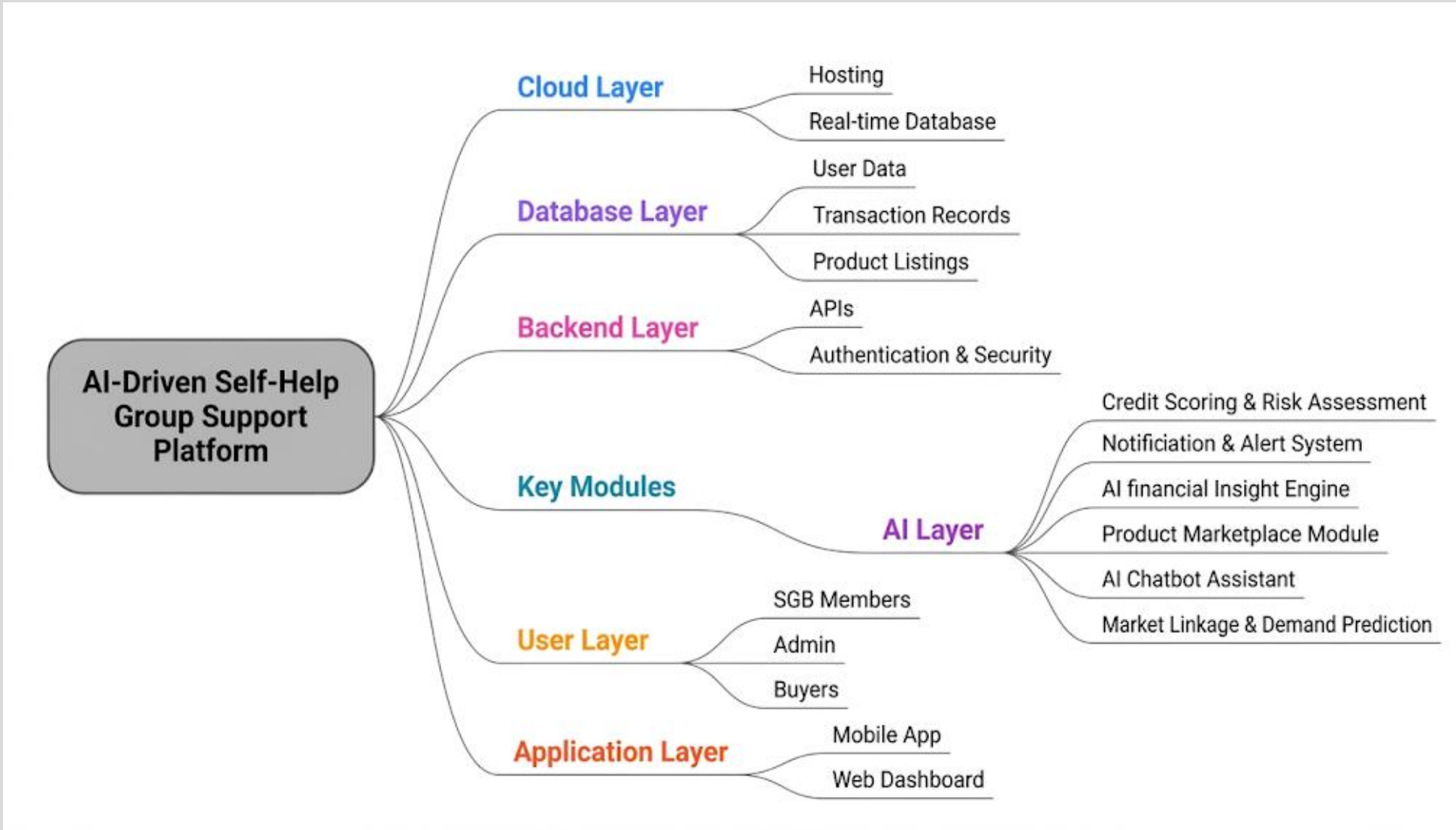


SHG Member's Perspective (The Solution: Digital Empowerment) — After

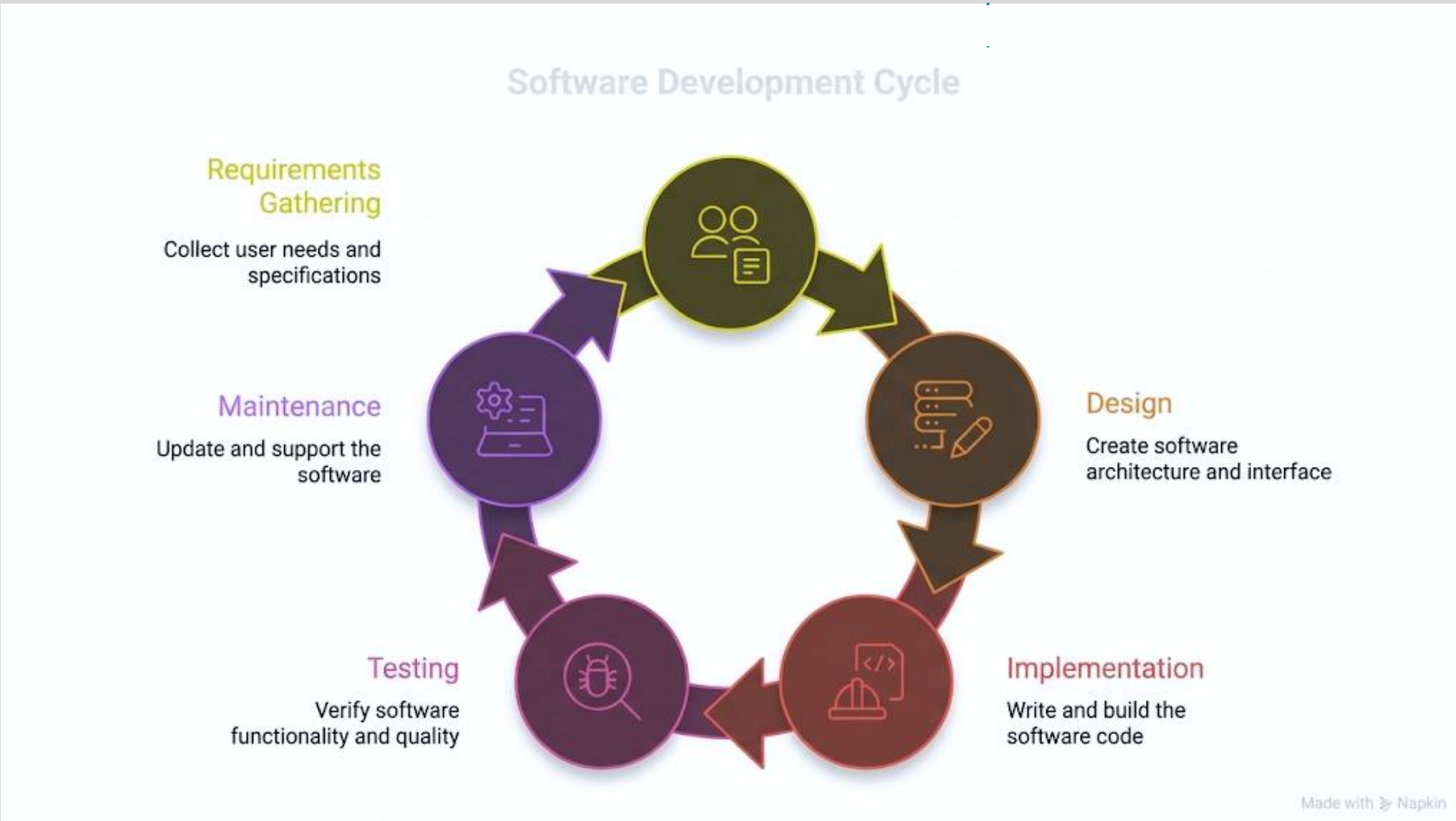


OVERALL FLOW DIAGRAM

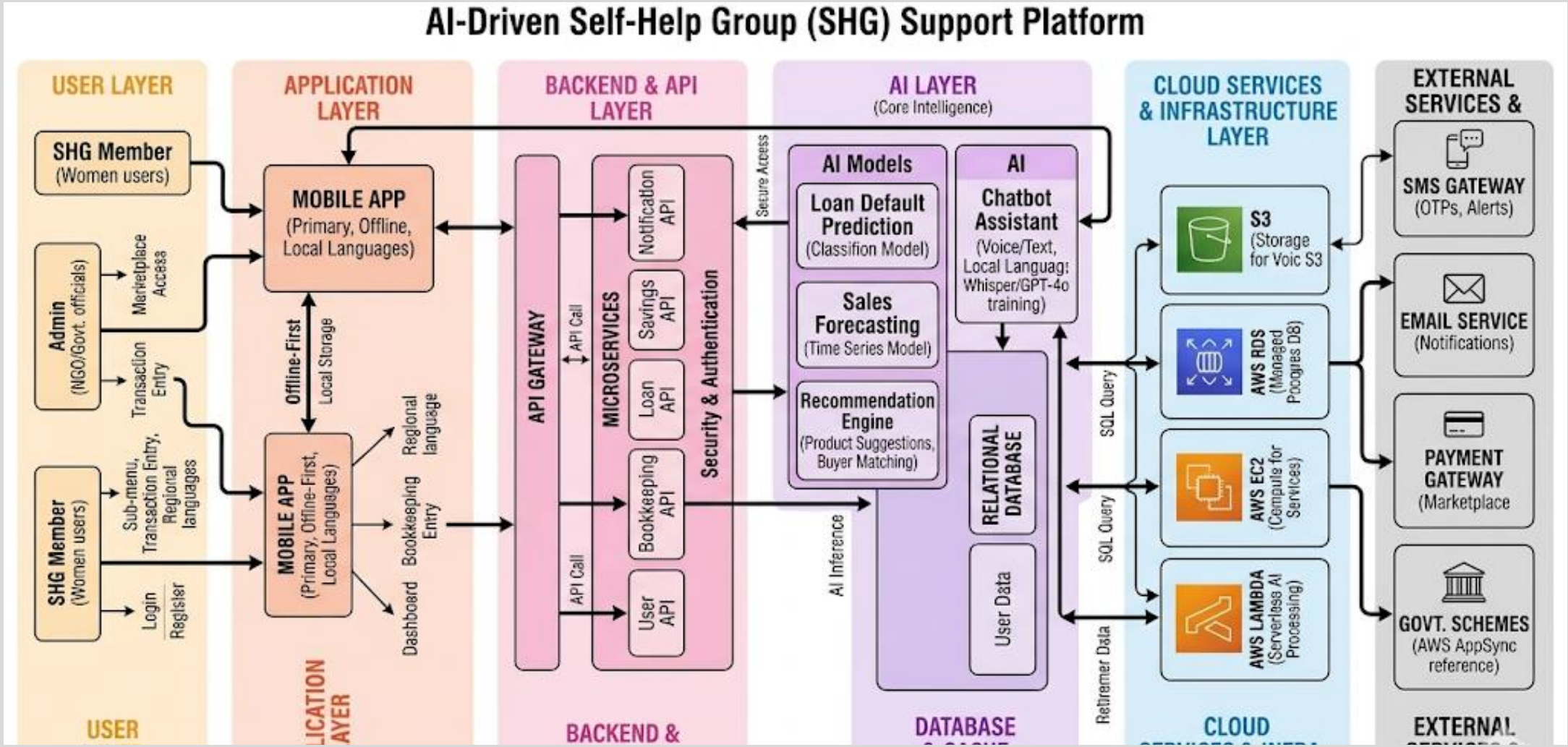
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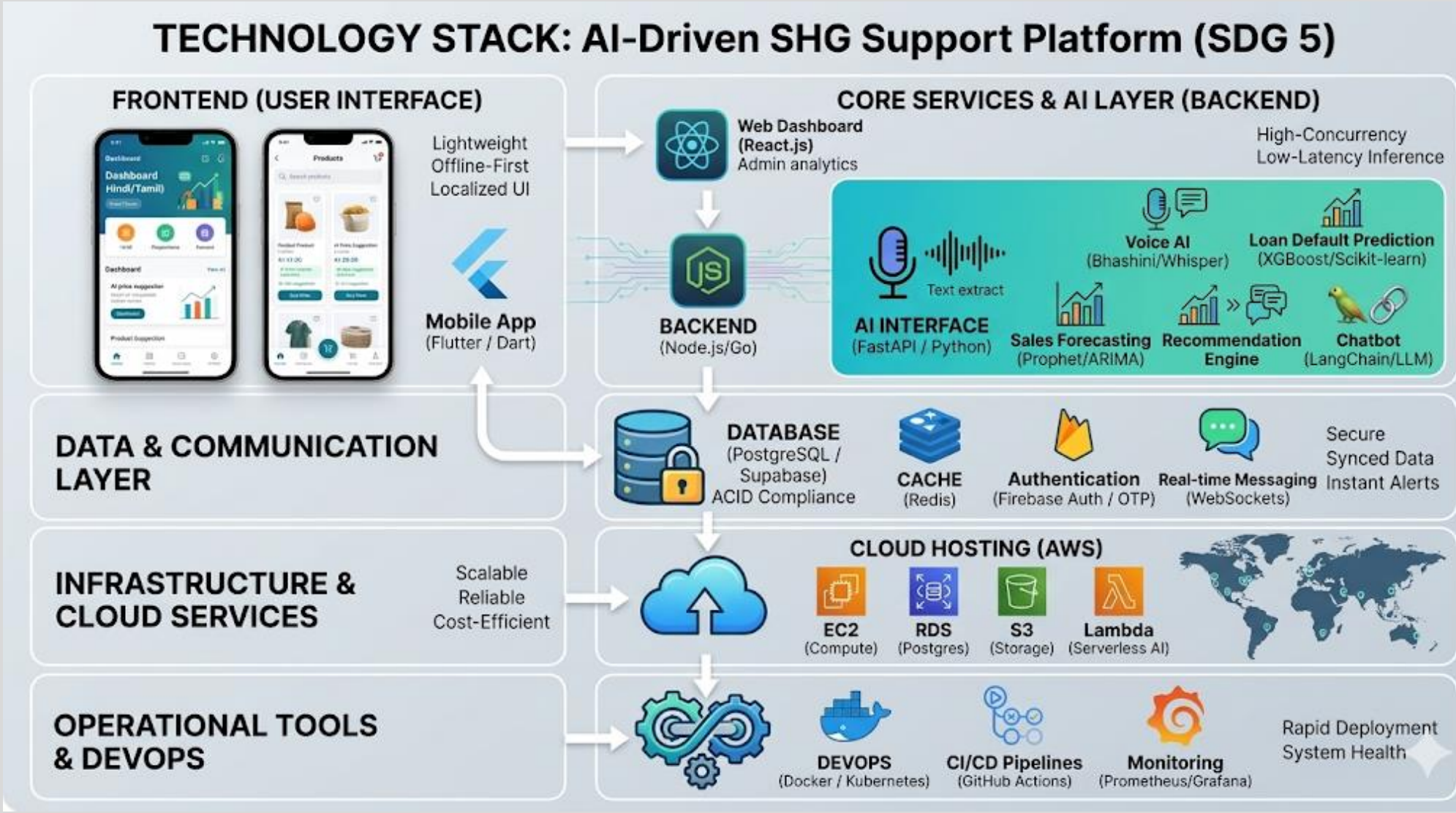
SOFTWARE FLOW DIAGRAM



BLOCK DIAGRAM



TECH STACK


TECHNOLOGY STACK: AI-Driven SHG Support Platform (SDG 5)
FRONTEND (USER INTERFACE)
Lightweight
Offline-First
Localized UI
Mobile App
(Flutter / Dart)
CORE SERVICES & AI LAYER (BACKEND)
Web Dashboard
(React.js)
Admin analytics
High-Concurrency
Low-Latency Inference
BACKEND
(Node.js/Go)
AI INTERFACE
(FastAPI / Python)
Text extract
Voice AI
(Bhashini/Whisper)
Loan Default Prediction
(XGBoost/Scikit-learn)
Sales Forecasting
(Prophet/ARIMA)
Recommendation
Engine
Chatbot
(LangChain/LLM)
DATA & COMMUNICATION
LAYER
DATABASE
(PostgreSQL /
Supabase)
ACID Compliance
CACHE
(Redis)
Authentication
(Firebase Auth / OTP)
Real-time Messaging
(WebSockets)
Secure
Synced Data
Instant Alerts
INFRASTRUCTURE &
CLOUD SERVICES
Scalable
Reliable
Cost-Efficient
CLOUD HOSTING (AWS)
EC2
(Compute)
RDS
(Postgres)
S3
(Storage)
Lambda
(Serverless AI)
OPERATIONAL TOOLS
& DEVOPS
DEVOPS
(Docker / Kubernetes)
CI/CD Pipelines
(GitHub Actions)
Monitoring
(Prometheus/Grafana)
Rapid Deployment
System Health

MODULE DESCRIPTION

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Smart Bookkeeping Module

This module is responsible for digitizing all financial activities of the Self-Help Group. It allows users to record savings, loans, repayments, and expenses through a simple mobile interface. The system automatically generates balance sheets and transaction summaries, reducing manual errors. It also supports voice-based data entry in local languages, making it easy for users with low literacy to use the system effectively

AI Financial Insights Module

This module uses machine learning algorithms to analyze financial data and provide useful insights. It predicts the risk of loan default, identifies irregular transactions, and suggests better savings plans for members. By analyzing past data, the system helps SHGs make informed financial decisions and improves transparency and accountability within the group.

Product Marketplace Module

Empowers SHGs to sell handicrafts and food directly to consumers, eliminating middlemen. Features include AI-driven pricing and enhanced visibility for wider market reach.

AI Chatbot Assistant

The chatbot acts as a virtual assistant for SHG members. It supports both text and voice interaction in regional languages like Tamil and Hindi. Users can ask questions related to loans, savings, government schemes, and business ideas. The chatbot provides instant, easy-to-understand responses, improving accessibility and user engagement.

Market Linkage & Demand Prediction Module

This module analyzes market data to identify current demand trends and suggests which products SHGs should focus on. It helps members understand what to produce, when to produce, and where to sell. The system can also recommend nearby markets or online platforms to maximize sales opportunities.

Notification & Alert Module

This module keeps users updated with important information through real-time alerts and notifications. It sends reminders for loan repayments, meeting schedules, and updates about product sales. This ensures that members stay informed and do not miss any important financial or group activities.

Direct Competitors

1. Aajeevika SHG App-Developed for SHGs under government programs Tracks group activities & records

2. e-SHG App - Digital bookkeeping for SHGs

3. Khatabook - Digital ledger for small businesses

What Makes OUR Project Unique?

An AI-powered, all-in-one platform built specifically for women SHGs that combines smart bookkeeping, loan risk prediction, and a direct product marketplace.

Unlike existing apps, it offers data-driven insights, local language support, and voice-based usage, making it accessible and practical for rural user



BUDGET & COST ESTIMATION

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COMPONENT	COST(INR)
DEVLOPMENT(STUDENT)	0-2000
CLOUD SERVICES	2000 - 10000
AI & API'S	1000 - 3000
VOICE FEATURES	1000 - 5000
DOMAIN & HOSTING	3000 - 5000
TESTING & MAINTENANCE	2000 - 5000

PROTOTYPE - 3K - 15K

SEMI PROFESSIONALS - 20K - 60K

COMMERCIAL - 1L - 3L

FUTURE STEPS

WADHWANI FOUNDATION

COMMUNITY OUTREACH



Spread awareness and promote app among SHGs

APP LAUNCH



Launch app in local languages for women SHGs

USER TRAINING & SUPPORT



Conduct training sessions & provide local language support

PARTNERSHIPS & COLLABORATION



Work with NGOs, banks & government schemes

MARKET ACCESS & E-COMMEERCE



Enable product listing & e-commerce for SHGs

AI & DATA ANALYTICS



Use AI to provide financial insights & risk assessments

FEEDBACK & IMPROVEMENTS



Gather user feedback, improve & add new features

MONITORING & GOVERNMENT COMPLIANCE



Continuously monitor app usage & follow government guidelines

EmpowerHER

AI-Powered Finance & Marketplace App

Empowering Women. **Elevating minds everywhere.**

WADHWANI FOUNDATION

Empowering Women. Elevating minds everywhere."

1. The Problem

- No digital financial management
- High loan defaults (40-50%)
- Limited market access
- Language barriers for rural women

There are **8M+** Women SHGs in India!



There are **8M+** Women SHGs in India!



2. Our Solution

EmpowerHER is an app to manage deports:

- Savings & loans:
- Sell products online
- **AI-based risk prediction**
  I'natare/text chat in SriGs

3. Market Opportunity

India SHG Finance Market:

$14 Billion+



6. Impact & SDGs

- **Premium Subscription Plan**
  ₹149 – ₹499/month
- **Small Transaction Fees**
  1–5% sales.commission:
- **Partnered Lending & Ads**
  Tte-ups with MFIs, government schiemes

4. Key Features

- **Loan Risk Prediction**
  Al warns default risk,
- **Marketplace & Payments**
  Sell products online
- **Voice/Text Chatbot**
  Ask questions in local language.

5. How It Works

- User inputs finance/product data
- App analyzes yia generate insights
- **Mobile internet** penetration
  Treriaity., 50G 11 = 8¥0 ttitls, user.

7. Revenue Streams

- **Premium Subscription Plan**
  ₹149 – ₹499/month
- **Small Transaction Fees**
  1–5% sales.commission)
- **Partnered Lending & Ads**
  The upe with MFIs, government schemes.

4. Key Features

- **Dashboard & Reports**
  Visualize savings/loans, serc
- **Product Recommendations**
  Al suggests popular Tterns.

5. Impact & SDGs

- **Premium Subscription Plan**
  ₹149 – ₹499/month:
- Small Transaction Fees
  1–5% sales commission

8. Technology Stack

- Mobile App: Flunter/React Native
- Backend Node.js & Python
- Al Models S,/kit-kearn, TensorFlow
- Database Firebase:



Our Vision: A future where SHGs thrive with financial independence and **community impact.**

CONCLUSION

The **AI-Driven Self-Help Group (SHG) Support Platform** is an **innovative solution** designed to **digitally transform** the way SHGs operate by replacing **manual processes** with an **intelligent, automated system.**

It helps women efficiently manage their **savings, loans,** and **financial records** while **reducing errors** and **improving transparency**.

By integrating **Artificial Intelligence (AI)**, the platform provides valuable insights such as **loan risk prediction, savings recommendations**, and demand analysis, enabling better financial planning and decision-making.









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Thank You!